METRIC

MIL-P-63245B (AR)
AMENDMENT 3
14 May 1998
SUPERSEDING
AMENDMENT 2
06 November 1996

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, M795
METAL PARTS ASSEMBLY

This amendment forms a part of MIL-P-63245B (AR) dated 16 October 1995 and is approved for use by the US Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 STANDARDS:

Delete: "MILITARY (Advisory)"

"MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification?

"MIL-STD-1949 - Inspection, Magnetic Particle"

PAGE 3

Non-Government publications:

Add: "E 399 - Standard Test Method for Plane-Strain Fracture Toughness of Metallic Materials

E 1444 - Magnetic Particle Testing

AMSC N/A FSC 1320
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AMERICAN NATIONAL STANDARDS INSTITUTE

CP 189 - Standard for Qualification and Certification of Nondestructive Testing Personnel”

3.7.1.2 Subsurface defects.

Change from:

“There shall be no evidence of either intergranular penetration of gilding metal or weld arc cratering except for the starting weld pass when the body band seat is turned down to 147.19 plus or minus 0.25 millimeters (mm) diameter (5.795 inches plus or minus 0.01 inches) and tested as specified in 4.4.3.2. There shall be no evidence of intergranular penetration of gilding metal or weld arc cratering when the body band seat is turned down to 144.65 plus or minus 0.25 millimeters (mm) diameter (5.695 inches plus or minus 0.01 inches) and tested as specified in 4.4.3.2.”

* Add:

“3.7.4 Protection of rotating band. The rotating band shall be protected after final machining from being nicked, burred, or otherwise damaged in subsequent handling prior to shipment.”

* 3.8.2 Rotating band.

Delete in its entirety and replace with the following:

“3.8.2.1 Rotating band. The surface of the rotating band lands shall be examined for surface defects, damage and contamination.

a. On the forward land, there shall be no evidence of raised copper due to nicks, digs or damage of any kind and there shall be no paint or foreign matter on the outer diameter.

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b. On the aft three lands, any nicks or digs extending 1/2 the land width or less are acceptable, provided the maximum band diameter is not exceeded.

c. The surface of the overlay band shall have no cavities that exceed 2.38mm (3/32 inch) in the greatest dimension.

d. On the surface of the overlay band the number of cavities equal to or less than 2.38mm (3/32 inch) in the greatest dimension shall not exceed one (1) per 45 degree segment of the band. However one (1) 645 square mm (1 square inch) area per band is allowed a maximum of four (4) cavities none of which exceeds 2.38mm (3/32 inch) in the largest dimension.

e. On the surface of the overlay band one tightly closed seam in the area of the weld start stop zone is allowed.

f. Disbond (pick out) as indicated by a ragged appearance forward or aft of the rotating band is allowed.

g. The overlay rotating band and band seat areas shall have no other discontinuities, cracks, lack of bonding, lack of fill or other metal defects other than those herein defined.

h. Severity of defects shall be in accordance with visual standards."

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* Slow bend precracked charpy test.

Delete in its entirety.

Replace with the following new paragraph:

"3.10 Plan-Strain Fracture Toughness test. The minimum fracture toughness (ic) in the side wall shall be 25 per square inch (thousand pounds square inch square root inch). The minimum fracture toughness (Kic) in the base shall be 22.5 KSI in. (thousand pounds per square inch square root inch)."

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PAGE 7

4.3.1.1 Submission:

Delete “Base Cover . . . . . 9326051 . . . . . 5”

PAGE 8

* Tests to be performed and acceptance criteria.

Delete: “4.5.9” and substitute: “4.5.7”

PAGE 9

* Table I First Article Inspection

Body Assembly

Add:

“Yield strength and elongation test 2 3.3 4.5.3/4.4.3.1
Plane-Strain Fracture Toughness test 1 3.10 4.5.3/4.4.3.1.4”

Body, Machined

Delete:

“Yield strength and elongation test 2 3.3 4.5.3./4.4.3.1
Slow bend precrack Charpy 1 3.10 4.5.3.3/4.4.3.1.4”

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Table I. First Article Inspection

Delete:

“Base Cover (Dwg. 9326051) Examination of characteristics
5 3.2 4.4.2.1”

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PAGE 11

4.4.1.1 Heat treatment inspection lot:

Change first sentence from:

"Heat treatment inspection lots shall not exceed one mill heat of steel or 200 components, whichever is smallest."

To:

"heat treatment inspection lots shall not exceed one mill heat of steel or 2500 components, whichever is smallest."

PAGE 14

4.4.2.1 Base Cover

Delete in its entirety.

PAGE 17

* Quality Conformance Inspection

4.4.2.4 Body Machined

Special

delete:

"12 Precrack Charpy test 4.4.3.1.3 3.10 4.5.3.3"

Major

delete:

"106 Yield strength and elongation of body

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* Quality Conformance Inspection

4.4.2.5 Body Assembly

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Special

11 Presence of base plate, unbroken base plate weld and unperforated base plate.

Delete in its entirety.

12 Ultrasonic test

Delete: "4.5.6" and substitute: "4.5.5"

* Add:

"16 Plane-Strain Fracture Toughness test 4.4.3.1.3 3.10 4.5.3.3"

* Quality Conformance Inspection

4.4.2.5 Body Assembly

Major

111 Absence of surface voids, seams, iron deposits or cracks in rotating band.

Delete in its entirety and replace with:

"111 Rotating band defective Level III 3.8.2.1 Visual'Probe/Scale"

Add:

"114 Yield strength and elongation of body 4.4.3.1 3.2 4.5.3

* 4.4.2.6 Projectile, HE, 155mm, M795 Metal Parts Assembly

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Delete: “104 Salt spray test 4.4.3.4 3.2 4.5.6.1”

and replace with:

“104 Salt spray test 4.4.3.3 3.2 4.5.6.1”

* Hardness.

Change from:

“Each final heat treated projectile from a heat treated lot shall be identified and subjected to a hardness test on the surface indicated. One (1) hardness reading shall be taken. The hardness reading obtained shall be used as the basis for selecting the tension test sample of paragraph 4.4.3.1.2.”

Change to:

“Each final heat treated projectile from a heat treated lot shall be identified and subjected to a hardness test on the surface one half (1/2) inch forward of the rotating band. One (1) hardness reading shall be taken. The hardness reading obtained shall be used as the basis for selecting the tension test sample of paragraph 4.4.3.1.2.”

* 4.4.3.1.3 Slow bend precracked charpy test.

Delete in its entirety and substitute:

“4.4.3.1.3 Plane-Strain Fracture Toughness test. Beginning with the first heat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated projectile from each heat treated lot shall be subjected to the Lane-Strain Fracture Toughness test. The projectile selected shall be of the highest hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Four specimens shall be tested. No individual fracture toughness

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From the side wall shall be less than 25 pounds per square inch. No individual fracture toughness from the base shall be less than 22.5 pounds per square inch. If one test is invalid, acceptance may be based on the fracture toughness of the valid sample. If any individual fracture toughness is below the minimum allowable, the heat treated lot represented shall be rejected, or subjected to retemper and retest, for all mechanical properties, if approved by the Government. After three consecutive heat treated lots have been accepted, the test shall be discontinued. The test shall be reinstituted, until three consecutive heat treatment lots have complied with the applicable requirements, if any of the following occur:

a. Steel is procured from a different producer, or from a substantially different process as determined by the Government.

b. Forgings are procured from a different producer.

c. A substantial change is made in the forging process or configuration, as determined by the Government.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process, as determined by the Government.

f. A change is made to the heat treatment metal parts configuration."

PAGE 24

4.4.3.2.1 Rotating band seat location and runout.

Delete "4.4.2.2" and substitute "4.4.2.4"

* 4.4.3.32.2 Welder qualification sampling.

Subparagraph c.

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Change from:

"Turn down band seat to the diameter specified in 3.7.1.2 and again visually inspect in accordance with 4.5.4. Any defect indication per 3.7.1.2 at this depth is considered to be a reject."

To:

"Turn down the band seat to the largest diameter specified in 3.7.1.2 and again visually inspect in accordance with 4.5.4. Any defect indication per 3.7.1.2 other than at the starting weld ;pass is considered to be a reject. If indications are present at the starting weld pass turn down the band seat to the smallest diameter specified in 3.7.1.2 and again visually inspect in accordance with 4.5.4. Any defect per 3.7.1.2 is considered a reject. The starting weld pass area is defined as ¾ inch maximum on each side of the start-stop weld center line."

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4.4.3.2.3.1 Etched bandseat examination. Subparagraph d.

Delete "4.4.3.2.2.1a, 4.4.3.2.21b, and 4.4.3.2.1" and substitute "4.4.3.2.3.1 a., 4.4.3.2.3.1 b., and 4.4.3.2.2"

4.4.3.2.3.2 Sub-surface defect examination.

Delete "4.4.3.2.1" and substitute "4.4.3.1.3"

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* 4.5.3.1 Hardness. Subparagraph a. first sentence change from:

"The surface of the body in the area designated by the drawing to be hardness tested shall be prepared by machining to a sufficient depth, 0.38 millimeters (.015 in.) minimum, to remove scale and get below the decarburization zone."

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Change to:

"The surface of the body in the area designated to be hardness tested shall be prepared by machining to a sufficient depth, 0.38 millimeters (.015 in.) minimum, to remove scale and get below the decarburization zone."

Subparagraph (3)

Delete "4.4.3.1.4" and substitute "4.4.3.1.3"

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* 4.45.3.3 Slow bend precrack charpy.

Delete in its entirety and replace with the following:

"4.5.3.3 The two base specimens shall be taken from locations F1 and F2 on Figure 1. The test shall be conducted at a specimen temperature of –54 degrees C (-65 degrees Fahrenheit). The test shall be conducted in accordance with the procedure defined with the procedure defined in ASTM E399. The notches shall be machined across the rear face of the base specimen. Locations F3 and F4 on Figure 1 shall be used for replacement specimens for locations F1 and F2 respectively. The two side wall specimens shall be taken from locations F5 and F6 on Figure 2. The test shall be conducted at a specimen temperature of –54 degrees C (-65 degrees Fahrenheit). The test shall be conducted in accordance with the procedure defined in ASTM E399. Orientation of the side wall samples shall be L-R with the notches machined across the outside diameter surface of the specimen. Locations 180 degrees from samples F5 and F6 shall be used for replacement specimens for locations F5 and F6 respectively."

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4.5.4 Magnetic particle inspection (Rotating band area):

Change first sentence from:

"The inspection shall be in accordance with MIL-STD-1949, or equivalent commercial standard as approved by the contracting officer, using the continuous, wet fluorescent method."

To:

"The inspection shall be in accordance with ASTM E1444, using the continuous, wet fluorescent method."

* Change next to last sentence from:

"The requirement of MIL-STD-1949 to check system effectiveness by means of steel ring specimen shall be waived."

To:

"The requirement of ASTM E1444 to check system effectiveness by means of steel ring specimen shall be waived."

* 4.5.4.1 Operator qualification.

Change From:

"Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I, or equivalent commercial standard as approved by the contracting officer. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with Mil-STD-410, Level II."

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To:

"Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with ANSI CP 189, Level I. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with ANSI CP 189, Level II."

4.5.5 Ultrasonic inspection.

Change from:

"Ultrasonic inspection. The projectiles shall be ultrasonically inspected using equipment and procedures in accordance with 4.5.5.1. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Standards, in accordance with 4.5.5.3, shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.8 shall be classed defective and removed form the lot. All projectiles rejected by ultrasonics shall be visually inspected for the location of the discontinuity. If the discontinuity can be observed visually, it shall be characterized as a critical defect, and appropriate actions shall be taken.

To:

"Ultrasonic inspection. The projectiles shall be ultrasonically inspected using equipment and procedures in accordance with 4.5.5.1. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Standards, in accordance with 4.5.5.3, shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.8 shall be classed defective and removed form the lot. All projectiles rejected by ultrasonics shall be visually inspected at the location of the ultrasonic indication. An ultrasonic critical defect shall be defined as nay ultrasonic reject that has a visual crack or pipe or a visual defect other than a crack or pipe

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(Lap, Fold, Seam, etc.) that has an ultrasonic signal that is twice the level of the ultrasonic standard notch for that area. Visual defects other than cracks or piping shall be statically evaluated by a level II inspector to determine if a critical defect exists. Critical defect determination shall be accomplished before that start of the next production shift."

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4.5.5.4 Operator qualification.

Change from:

"Operator qualification. Personnel performing the ultrasonic inspection shall be certified in accordance with MIL-STD-410 or equivalent commercial standard as approved by the contracting officer, with the following exception:"

To:

"Operational qualification. Personnel performing the ultrasonic inspection shall be certified in accordance with ANSI CP-189, with the following exception:"

PAGE 32

4.5.6.1 Salt spray.

Delete "4.4.5.1" and substitute "4.4.3.3"

PAGE 34

6.6 Ballistic performance test summary.

Under heading "Sample size"

Delete "(see 4.4.5.1)" and substitute "(see 4.4.5)"

6.10 Data requirements.

Delete "4.5.2" and "4.5.3" and substitute "4.5.3.1" and "4.5.5.2" respectively.

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Delete “Test Report” and substitute “Test Equipment and Procedures”

Delete “DI-T-1906” and substitute “DI-R-1714”

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* 6.13 Band welding.

Change from:

“It is recommended that the welding operation of the overlay welded band be guided by Frankfort Arsenal “Process recommendations for Banding 155MM, M483 Projectile” dated 19 April 1973. Requests for copies should be submitted to: Commander, ARDEC, ATTN: AMSTA-AR-CCH, Picatinny Arsenal, NJ 07806-5000.”

Change to:

“It is recommended that the welding operation of the overlay welded band be guided by Rotating Band Welding Procedure for M795 drawing 12982985.”

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* Appendix

Delete appendix in its entirety.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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